

Figure 1: Limitations of the standard secant condition. (a) Function values and gradients at  $x_k$  and  $x_{k+1}$  are already known. (b) The ideal quadratic model constructed from the exact Hessian fits well around the new point  $x_{k+1}$ . (c) The standard secant condition may fail to capture curvature adequately.

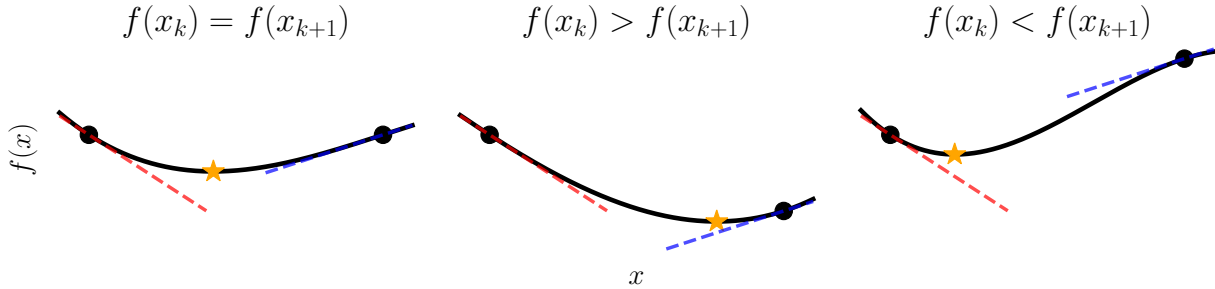


Figure 2: Basic idea of the modified secant condition. Even when the gradients are the same at two points  $x_k$  and  $x_{k+1}$ , natural interpolation of function values differs depending on the function values  $f(x_k)$  and  $f(x_{k+1})$ .

## 1 Modified Secant Condition

This section explains the modified secant condition, a modification of the standard secant condition. The modified secant condition incorporates function value information in addition to gradient information, leading to more accurate Hessian approximations.

### 1.1 Derivation of the Modified Secant Condition

Although the secant condition is widely used in many quasi-Newton methods, it can fail to capture the curvature of the objective function accurately because it only utilizes gradient information. We illustrate this limitation in Figure 1. In Figure 1, we have two points  $x_k$  and  $x_{k+1}$  where function values and gradients are already known. We see that the ideal quadratic model constructed from the exact Hessian fits well around the new point  $x_{k+1}$ , leading to a better convergence behavior. However, the standard secant condition uses only gradient information and ignores function values. Consequently, it may significantly misestimate curvature, leading to a poor approximation of the true objective function.

We can overcome this limitation by utilizing the function values. The basic idea is illustrated in Figure 2. Even when the gradients are the same at two points  $x_k$  and  $x_{k+1}$ , natural interpolation of function values differs depending on the function values  $f(x_k)$  and  $f(x_{k+1})$ . This observation motivates the modified secant condition, which incorporates function value information to improve Hessian approximation.

In the following, we present two well-known modified secant conditions.

### 1.1.1 Function-Value-Matching Modified Secant Condition

The first modification enforces the quadratic model to match the function value at the previous point [1, 4, 6]. Let us consider an alternate model with a different approximate Hessian  $B_{k+1}^F$ :

$$m_{k+1}^F(x) := f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2}(x - x_{k+1})^\top B_{k+1}^F (x - x_{k+1}).$$

We require this model to satisfy

$$m_{k+1}^F(x_k) = f(x_k),$$

which enforces that the function value at the previous point is correctly modeled.

Let us derive the corresponding modified secant condition for  $B_{k+1}^F$ . Substituting the definition of  $m_{k+1}^F$  and using  $s_k = x_{k+1} - x_k$ , the condition becomes

$$\begin{aligned} f(x_k) &= f(x_{k+1}) + \nabla f(x_{k+1})^\top (x_k - x_{k+1}) + \frac{1}{2}(x_k - x_{k+1})^\top B_{k+1}^F (x_k - x_{k+1}) \\ &= f(x_{k+1}) - \nabla f(x_{k+1})^\top s_k + \frac{1}{2}s_k^\top B_{k+1}^F s_k. \end{aligned}$$

Now, instead of the standard secant condition, we assume a modified form with an additional term involving the identity matrix:

$$(B_{k+1}^F + \sigma_k^F I) s_k = y_k.$$

Here,  $\sigma_k^F \in \mathbb{R}$  is a scalar to be determined as follows. Substituting this equation yields

$$f(x_k) = f(x_{k+1}) - \nabla f(x_{k+1})^\top s_k + \frac{1}{2}s_k^\top y_k - \frac{\sigma_k^F}{2} \|s_k\|^2.$$

Solving for  $\sigma_k^F$  and using  $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$ , we obtain

$$\begin{aligned} \sigma_k^F &= \frac{2(f(x_{k+1}) - f(x_k)) - (2\nabla f(x_{k+1}) - y_k)^\top s_k}{\|s_k\|^2} \\ &= \frac{2(f(x_{k+1}) - f(x_k)) - (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k}{\|s_k\|^2}. \end{aligned}$$

Therefore, the modified secant condition becomes

$$B_{k+1}^F s_k = \hat{y}_k^F := y_k + \frac{2(f(x_k) - f(x_{k+1})) + (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k}{\|s_k\|^2} s_k.$$

By using this modified  $\hat{y}_k^F$  instead of  $y_k$  in various quasi-Newton updates, we can incorporate function value information while keeping almost the same algorithm. Additionally, since condition  $s_k^\top y_k > 0$  is important for maintaining positive definiteness in BFGS update, we can consider the following modification to ensure that the inner product with  $s_k$  remains positive:

$$B_{k+1}^{F'} s_k = y_k + \frac{\max(0, 2(f(x_k) - f(x_{k+1})) + (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k)}{\|s_k\|^2} s_k.$$

This is the function-value-matching modified secant condition. Refer to Figure 3 for an illustration.

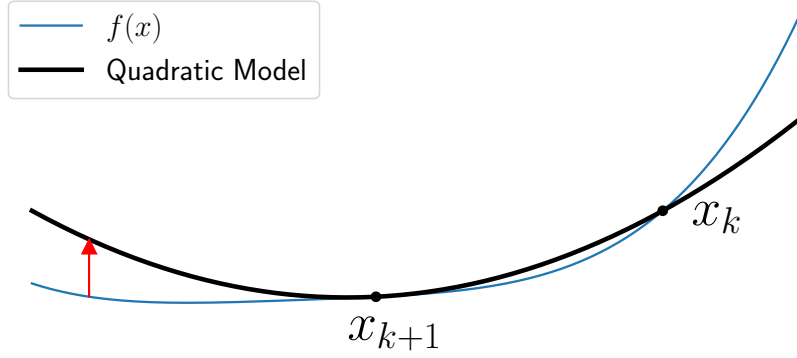


Figure 3: Function-value-matching modified secant equation. The function values  $f(x_k)$  and  $m_{k+1}^F(x_k)$  are matched at the previous point  $x_k$ .

### 1.1.2 Cubic-Augmented Modified Secant Condition

The second modified secant condition introduces a cubic term to the model, allowing simultaneous satisfaction of both the function value and the gradient matching at the previous point [5, 7, 8]. Let  $T_{k+1} \in \mathbb{R}^{n \times n \times n}$  be the third-order derivative tensor of  $f$  at  $x_{k+1}$  such that

$$s_k^\top (T_{k+1} s_k) s_k = \sum_{i,j,l=1}^n \partial_{ijl} f(x_{k+1}) s_k^{(i)} s_k^{(j)} s_k^{(l)}.$$

Here,  $\partial_{ijl} f$  denotes the third derivative of  $f$  with respect to  $x_i$ ,  $x_j$ , and  $x_l$ , and  $s_k^{(i)}$  is the  $i$ -th component of the vector  $s_k$ . We introduced this tensor solely for analysis purposes, and it will be eliminated from the final formula.

By incorporating this tensor term, we can define the following model:

$$\begin{aligned} m_{k+1}^C(x) := & f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2} (x - x_{k+1})^\top B_{k+1}^C (x - x_{k+1}) \\ & + \frac{1}{6} (x - x_{k+1})^\top (T_{k+1} (x - x_{k+1})) (x - x_{k+1}). \end{aligned}$$

With this model, we can enforce both function value and gradient matching at the previous point  $x_k$ . Specifically, we require

$$\begin{cases} m_{k+1}^C(x_k) = f(x_k), \\ \nabla m_{k+1}^C(x_k) = \nabla f(x_k). \end{cases}$$

By substituting the definition of  $m_{k+1}^C$  and using  $s_k = x_{k+1} - x_k$ , we can rewrite these conditions as

$$\begin{aligned} f(x_k) &= f(x_{k+1}) - s_k^\top \nabla f(x_{k+1}) + \frac{1}{2} s_k^\top B_{k+1}^C s_k - \frac{1}{6} s_k^\top (T_{k+1} s_k) s_k, \\ \nabla f(x_k) &= \nabla f(x_{k+1}) - B_{k+1}^C s_k + \frac{1}{2} (T_{k+1} s_k) s_k. \end{aligned}$$

Rearranging and multiplying both sides by 3 for the first equation, and taking the inner product with  $s_k$  for the second equation, we have

$$3(f(x_k) - f(x_{k+1})) + 3s_k^\top \nabla f(x_{k+1}) = \frac{3}{2} s_k^\top B_{k+1}^C s_k - \frac{1}{2} s_k^\top (T_{k+1} s_k) s_k,$$

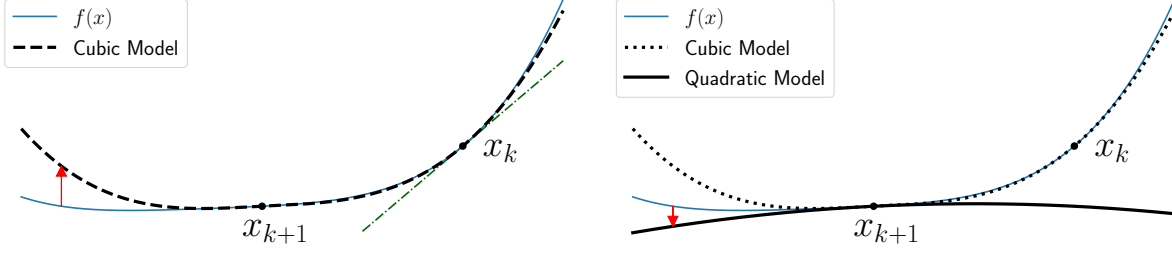


Figure 4: Cubic-augmented modified secant equation. The cubic term is incorporated into the model to simultaneously match both function value and gradient at the previous point. The model is constructed using the expansion up to the quadratic term.

$$-s_k^\top y_k = -s_k^\top B_{k+1}^C s_k + \frac{1}{2} s_k^\top (T_{k+1} s_k) s_k.$$

By adding these and using  $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$ , we eliminate the tensor term and obtain the scalar identity

$$3(f(x_k) - f(x_{k+1})) + \frac{3}{2} s_k^\top (\nabla f(x_{k+1}) + \nabla f(x_k)) + \frac{1}{2} s_k^\top y_k = \frac{1}{2} s_k^\top B_{k+1}^C s_k.$$

Again, for some scalar  $\sigma_k^C \in \mathbb{R}$ , we assume  $(B_{k+1}^C + \sigma_k^C I) s_k = y_k$ . Then, the previous equation yields

$$3(f(x_k) - f(x_{k+1})) + \frac{3}{2} s_k^\top (\nabla f(x_{k+1}) + \nabla f(x_k)) = -\frac{\sigma_k^C}{2} \|s_k\|^2,$$

which means

$$\sigma_k^C = -\frac{6(f(x_k) - f(x_{k+1})) + 3s_k^\top (\nabla f(x_k) + \nabla f(x_{k+1}))}{\|s_k\|^2}.$$

Thus, the modified secant condition becomes

$$B_{k+1}^C s_k = y_k + \frac{6(f(x_k) - f(x_{k+1})) + 3s_k^\top (\nabla f(x_k) + \nabla f(x_{k+1}))}{\|s_k\|^2} s_k.$$

This is a cubic-augmented modified secant condition. Refer to Figure 4 for an illustration.

## 1.2 Other Curvature Related Methods

There are several other topics related to obtaining the curvature pairs  $s_k, y_k$ . Agg-BFGS [2] is an alternative approach to managing curvature information by aggregating data rather than discarding the oldest information and adding the newest. Multi-Secant [3] extends the secant condition framework by maintaining multiple pairs of step and gradient difference vectors. In standard form we have seen, it is defined as

$$s_i = x_{i+1} - x_i, \quad y_i = \nabla f(x_{i+1}) - \nabla f(x_i). \quad (i = k-m, \dots, k)$$

In contrast, Multi-Secant centers all vectors around the latest iterate:

$$s_i = x_{k+1} - x_i, \quad y_i = \nabla f(x_{k+1}) - \nabla f(x_i). \quad (i = k-m, \dots, k)$$

This approach can lead to better approximations in some cases.

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